

Medical Examiner Office Information System

Author: Jeremy Dowling

Date: April 23, 2026

Role: Developer & Project Manager

Summary

Built a new module for the LabLynx LIMS (Laboratory Information Management System) to handle information management for the second largest Medical Examiner in the US, which remains in use to this day. Other state and county organizations have since adopted this software. All departments in the Medical Examiner office utilize the software — intake, investigations, medical records, X-Ray techs, pathology, tox lab, photography, etc.

The system includes full auditing and electronic signatures, and is routinely used in court proceedings. I handled all aspects of the project, including implementation, project management, design, requirements gathering, and the majority of development work. An independent web portal allows the public to make information requests. Automated features were implemented to support sub-contracted testing, including HL7 data exchange.

The go-live process included on-site training and manuals tailored to the Medical Examiner's processes.

Key Accomplishments

- Replaced a paper-based system — where some data was transcribed to a mainframe — with a fully electronic solution.
 - Migrated historical mainframe data going back to 1980 into the new software.
 - The Medical Examiner successfully demonstrated NAMEC and ISO compliance.
 - Automated HL7-based data and report exchange for sub-contracted testing.
 - Delivered a variety of worklists to facilitate status and prioritization tracking across all departments.
-

Technology Stack

Core Platform

Component	Technology
LIMS Platform	LabLynx LIMS (custom module)
Database	SQL Server
Web Server	IIS
Backend	ASP.NET, C#
Frontend	JavaScript, HTML
Reporting	GrapeCity ActiveReports

Automation & Integration

Component	Technology
Scripting	Java, RhinoScript, JavaScript
HL7 Integration	Mirth Connect
Workflow Automation	Node-RED

Screenshots

The screenshot displays the LIMS Admin - Medical Examiner interface. The top navigation bar includes 'About', 'Alert', and 'LogOut'. The left sidebar shows a 'Main Menu' with various options like 'First Call Sheet', 'ME Investigator Assn', 'ME Pathologist Assn', 'ME Cases', 'Cremation / Storage', 'Mass Disasters', 'Agencies', 'Agency Contacts', 'NME Queue', 'Indigent Queue', 'Unidentified Remains', 'Proficiency Samples', 'Tox Orders', 'New Lab Batch', 'Sample Batching', 'Data Entry', 'Sample Backlog', 'Search', 'Gift of Hope', 'Clinical Calculators', 'ICD-9 Lookup', 'ICD-10 Lookup', 'Consultation Requests', 'Addl Inv Requests', 'Tox Case Mgmt', 'Histology / Other', 'Reports and Labels', 'Pathology Dashboard', 'Document Management', 'Storage Locations', 'COC/Tracking Form', 'COC / Tracking Details', 'Body Release', 'Inventory Management', 'Requests', 'Internal Request Worklist', 'Travel Log', 'Corrective Action', 'Case Receipt Details', and 'IMEO Administration'.

The main form area is titled 'MEO/Add/Edit Case' and shows details for Case Number: ME2024-00010. The case status is 'Open' and the priority is 'Standard'. The name is 'Dumpty, Humpty' and the age/race/sex is '0/'. The tox status is 'No Samples' and the location is 'System Administrator'.

The form includes tabs for 'Records', 'Investigations', 'Intake', 'Triage', 'Pathology', 'Photography', and 'Done'. The 'Investigations' tab is active, showing the case details and a list of records.

The case details include:

- Case Number: ME2024-00010
- Location/Department: Medical Examiner/Entire Location
- Agency Name: LUTHERAN GENERAL HOSPITAL
- Agency Incident #: [Field]
- Priority: Standard
- Info Recd By: Admin, LIMS
- Date Info Recd: 10/30/2024 10:08
- Received By: Admin, LIMS
- Received Date: 10/30/2024 10:08
- Closed By: N/A
- Closed Date: [Field]
- Autopsy Rpt Cost: [Field]
- Protective Order Dt: [Field]
- Protective Order: [Field]
- Protective Order Exp Dt: [Field]
- Protective Order Contact: [Field]
- Initial Impression: Had a great fall

The 'Decedent' section includes:

- First Name: Humpty
- Middle Name: [Field]
- Last Name: Dumpty
- Individual Type: Deceased
- Gender: No Selection
- SSN: 000-00-0000
- Date of Birth (DOB): 11/24/2025
- Date of Death: 12/19/2025 04:46
- Birth Place: [Field]
- Citizenship: [Field]
- Marital Status: No Selection
- Industry: [Field]
- Veteran: No Selection
- Military: [Field]
- Yrs Elementary: [Field]
- Yrs College: [Field]
- Nationality: Select
- Latino: [Field]

The 'Age' section includes:

- Age: 0
- Age (Mos): 0
- Age (Dys): 25
- Age (hrs): [Field]

The 'Address' section includes:

- Address: [Field]
- City: Select
- County: Select
- State: No Selection
- Zip Code: [Field]
- Country: No Selection
- Area Code: [Field]
- Phone No: [Field]
- Phone #2: [Field]

Case Management Example